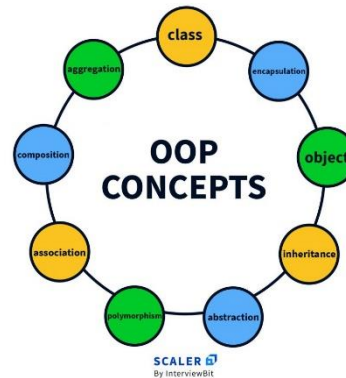


Object-Oriented Programming

OOP stands for Object-Oriented Programming, which is a programming paradigm that uses objects – instances of classes – for organizing code. It is based on several principles that help in designing and structuring code in a more modular and reusable way.

Following are the major characteristics for considering any programming language to be object oriented:

1. Class
2. Object
3. Data Abstraction
4. Data Encapsulation
5. Data Hiding
6. Inheritance
7. Polymorphism
8. Dynamic Binding
9. Overloading
10. Message Passing



1. CLASS: A class is a set of objects with similar properties, common behaviour, and common link to another object. The complete set data and common code of an object can be made a user defined data type with the help of class.

2. OBJECT: Object are important run time entities in object-oriented programming method. Each object contains data and code to manipulate the data.

3. DATA ABSTRACTION: Data Abstraction refers to the act of representing important description (action) without including the background details or explanation. C++ classes use the concept of abstraction and are defined as Abstract Data Type.

4. DATA ENCAPSULATION: Data Encapsulation means wrapping of data and function into a single unit called class. The data is not easy to access from the outside world and only that functions which are enclose in the class can access.

5. INFORMATION HIDING: Information Hiding means that the implementation detail of an object state and behaviour are hidden from users and other object to protect the state and behaviour unauthorized access.

6. INHERITANCE: Inheritance is the process of forming a new class from existing class or base class or super class or parent class. The class that is formed is called derived class or sub class or child class. Sub class inherit all the properties of their super class.

7. POLYMORPHISM: POLY means many and MORPH means form. Polymorphism means ability to take more than one form. Polymorphism play a main role in allocate objects having different internal structure to share the same external interface.

Example:

```
void area ( int ) ;  
void area ( int , int ) ;  
void area ( int ,int , int ) ;
```

8. DYNAMIC BINDING: Binding refers to linking of a function call to the code to be executed in response to the call.

9. OVERLOADING: Overloading is a type of Polymorphism. It allows an object to have different meaning depending on its context.

10. MESSAGE PASSING: In OOP, processing is accomplished by sending message to object. How message is executed is defined in the class function.

Advantages of OOP programming:

- a. Reuse of Code.
- b. Ease of Comprehension.
- c. Ease of fabrication and maintenance.
- d. Easy redesign and extension.

The difference between object-oriented programming and procedural programming.

SN	Object Oriented Programming	Procedural Programming
1	It follows bottom-up approach	It follows top up approach
2	Class and object-based abstraction	Procedural (function) based abstraction
3	by default, all the data are private	by default, all the data are public
4	Here emphasis is on object.	Here, emphasis is on doing things. (function)
5	Its feature like, data abstraction, encapsulation, polymorphism, inheritance are present.	Its feature like, data abstraction, encapsulation, polymorphism, inheritance are not present.
6	Code reusability is possible.	Code reusability is not possible.